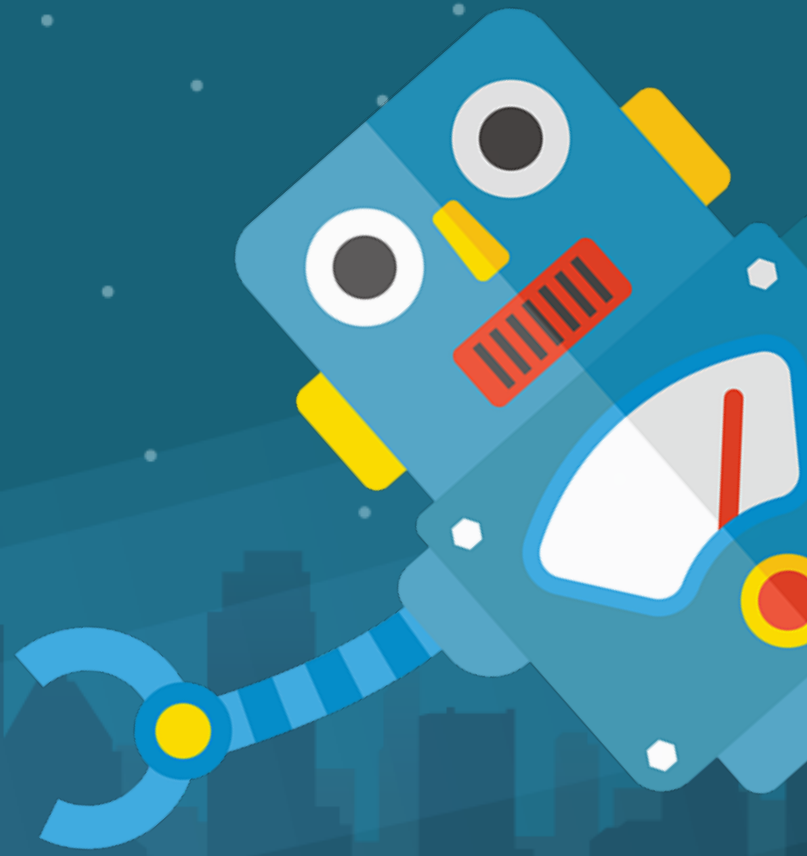
TECHORAMA

DEEP KNOWLEDGE IT CONFERENCE

October 1-3 | 2018

Ede, The Netherlands



XAMARIN: NEW AND NEXT



TECHORAMA

DEEP KNOWLEDGE IT CONFERENCE

October 1-3 | 2018 Ede, The Netherlands



David Ortinau

@davidortinau

21 yrs. web, interactive, mobile

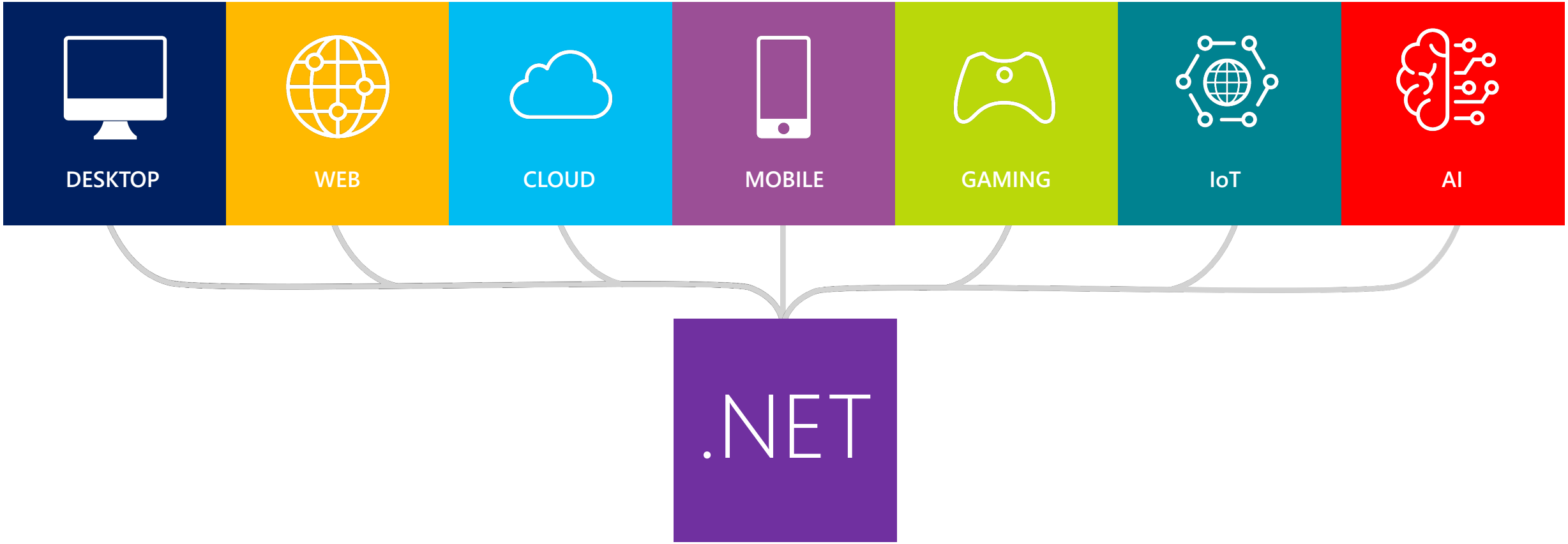
Xamarin MVP 2013 - 2017

BA English, Maryville University

Senior Program Manager
Visual Studio Mobile Developer Tools
Xamarin

Microsoft

Your platform for building **anything**

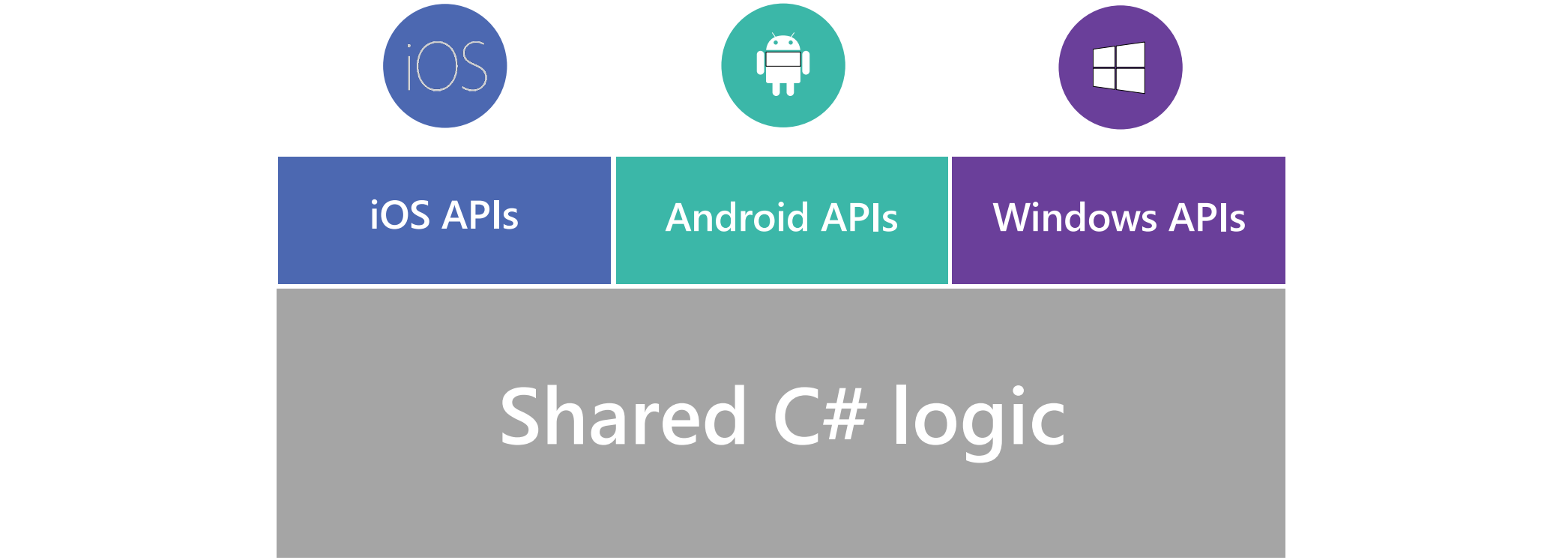


- Native user interface
- High-fidelity API access
- Native performance



Xamarin: Anything you can do in Objective-C, Swift, or Java, you can do in C# and Visual Studio

.NET for Mobile Apps



Shared C# codebase • 100% native API access • High performance

Xamarin Native approach

iOS C# | Android C# | Windows C#

Shared C# Logic

3 Native User Interfaces

Shared App Logic



Xamarin.Forms - Cross Platform Native UI





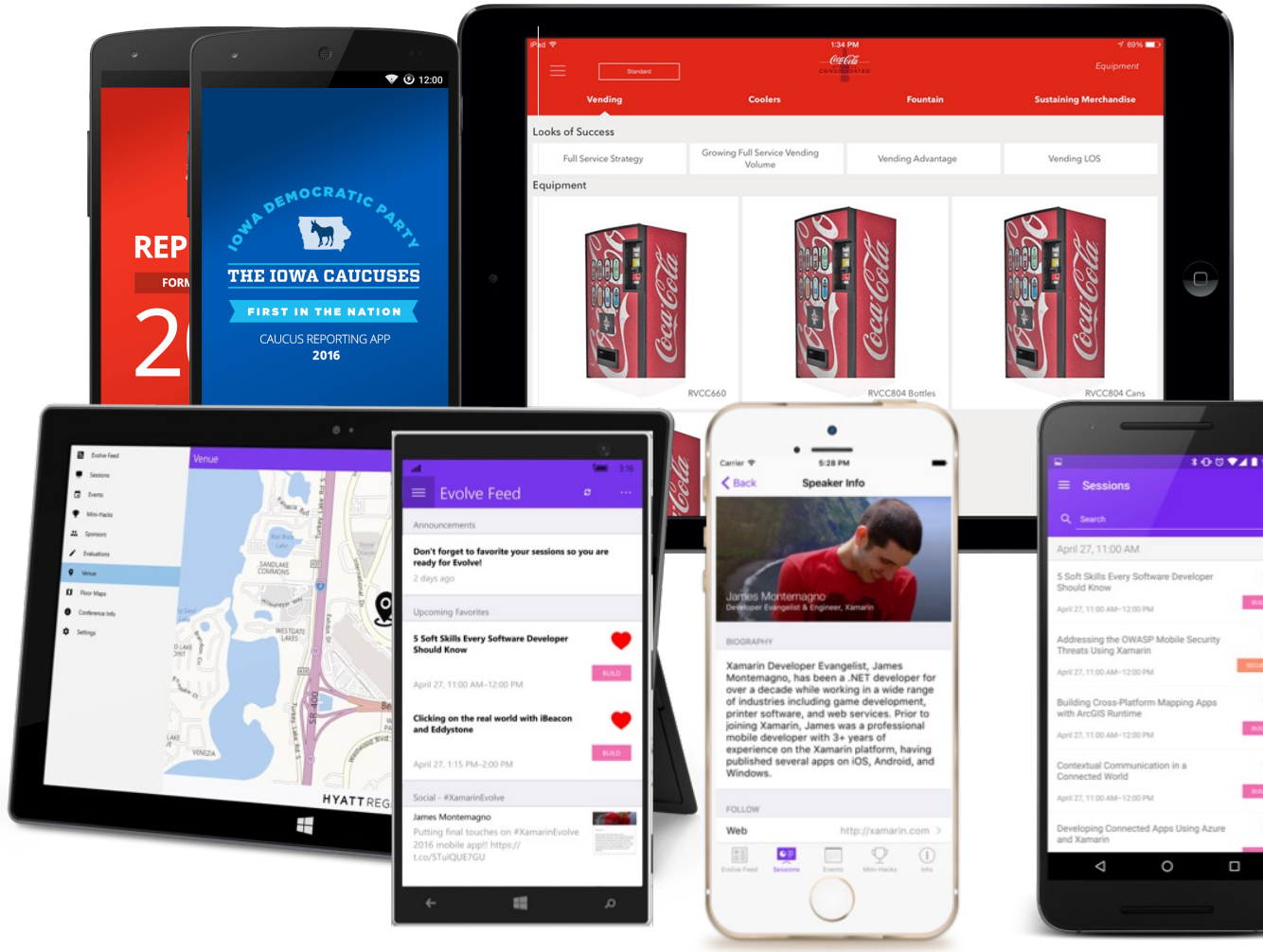


Shared UI Code

Shared C# Logic

Shared Native User Interface

Shared App Logic



Pages



Content



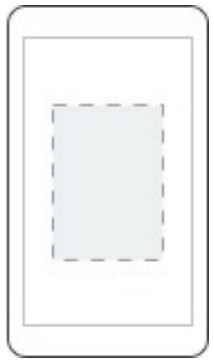
MasterDetail



Navigation

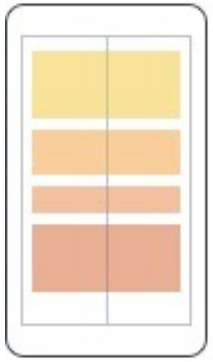


Tabbed

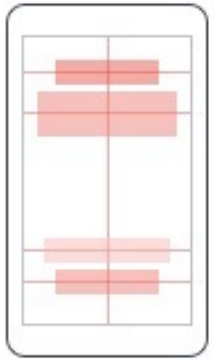


Carousel

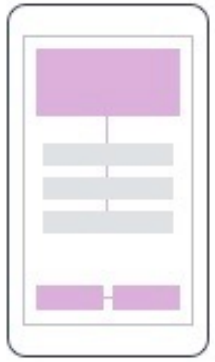
Layouts



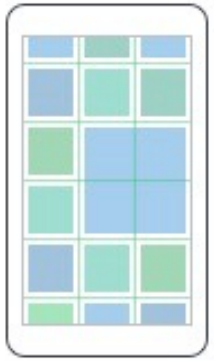
Stack



Absolute



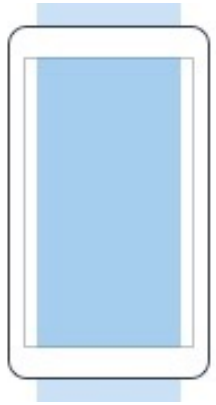
Relative



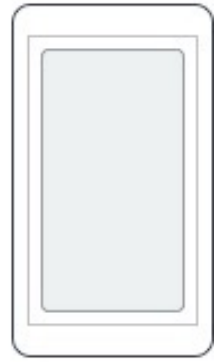
Grid



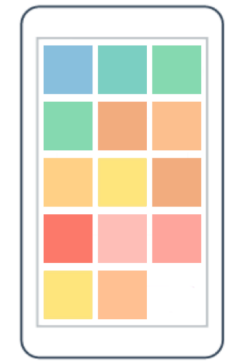
ContentView



ScrollView



Frame



Flex

Controls

ActivityIndicator	BoxView	Button	DatePicker	Editor
Entry	Image	Label	ListView	Map
OpenGLView	Picker	ProgressBar	SearchBar	Slider
Stepper	TableView	TimePicker	WebView	EntryCell
ImageCell	SwitchCell	TextCell	ViewCell	

Native UI

Forms elements are realized via platform renderers.

	ContentPage	Label	Picker
Android	Activity / Fragment	TextView	AlertDialog + LinearLayout + EditText + NumberPicker
iOS	UIViewController	UILabel	UITextField + UIPickerView + UIToolbar + UIBarButtonItem
UWP	Page	TextBlock	ComboBox

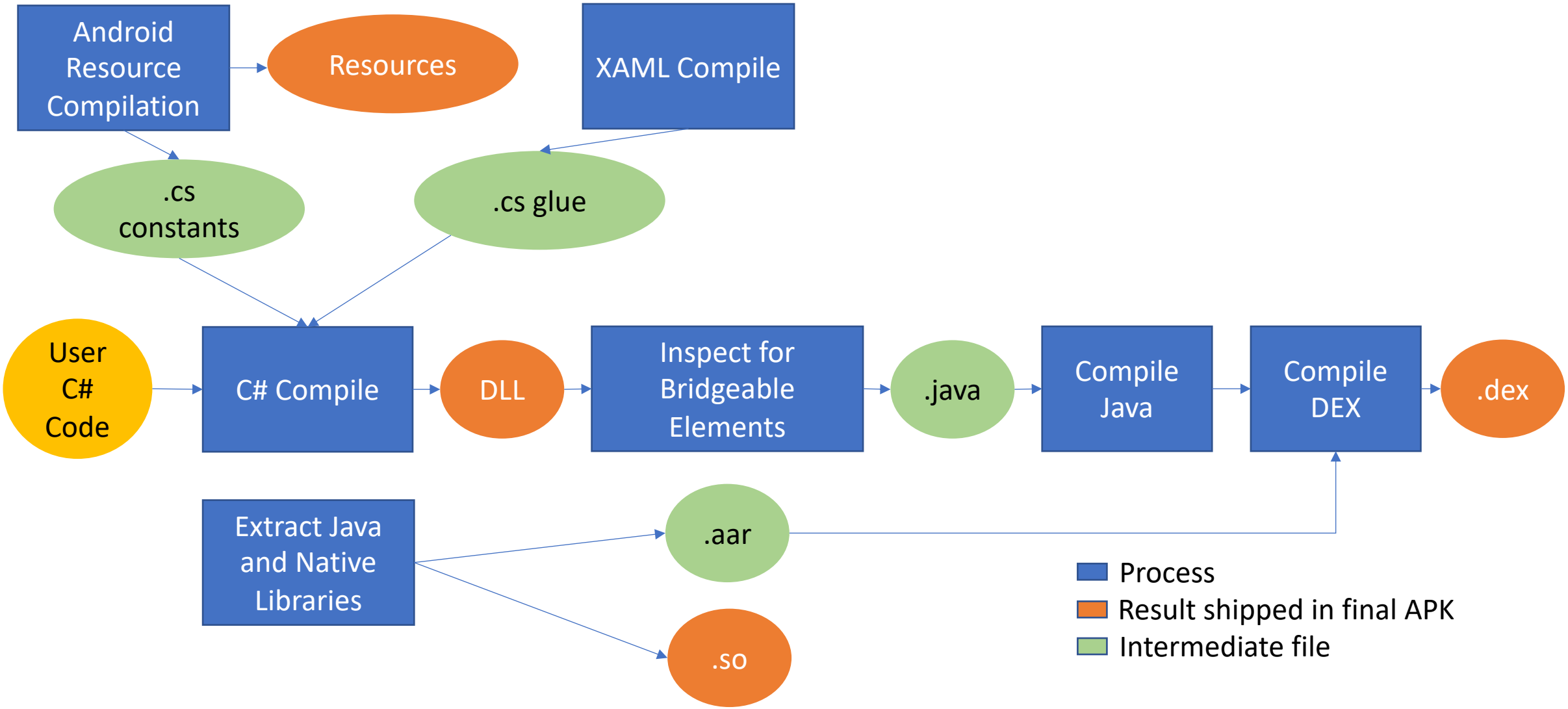
developer.xamarin.com/guides/xamarin-forms/application-fundamentals/custom-renderer/renderers/



Improving Build Times

- Optimizing full builds
  - Optimize passes
  - Fix dependencies
  - Avoid unnecessary rebuilds
- Optimize incremental builds
  - For your continuous testing

Creating an Android Package (APK)

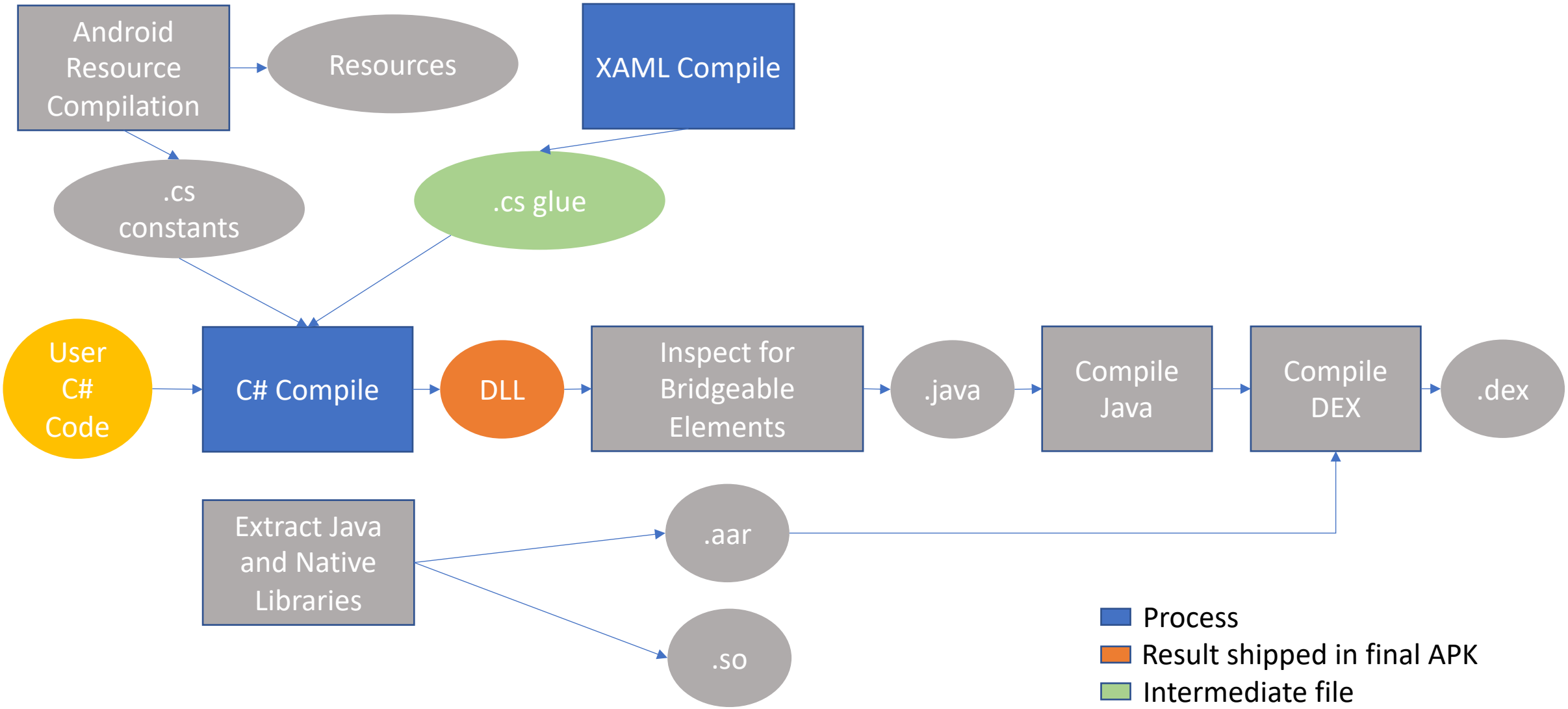


Cumulative Results – Complex app

- Smart Hotel 360 application
  - 25,000 lines of C#
  - 99 NuGet packages
    - 47 from .NET standard
    - 16 Android Support Libraries (“extras”)
    - 5 Google Play Services
    - 3 Azure
    - 26 other (Skia + native code, Splat, Plugins, NewtonSoft.Json, helpers)
  - 32 XAML screens

- Stable release (15.9): 02:15s
- Current master: 0:57s

Fast Forms Inner Loop



Speeding up your Workflow

- Rebuild and run in 3 seconds

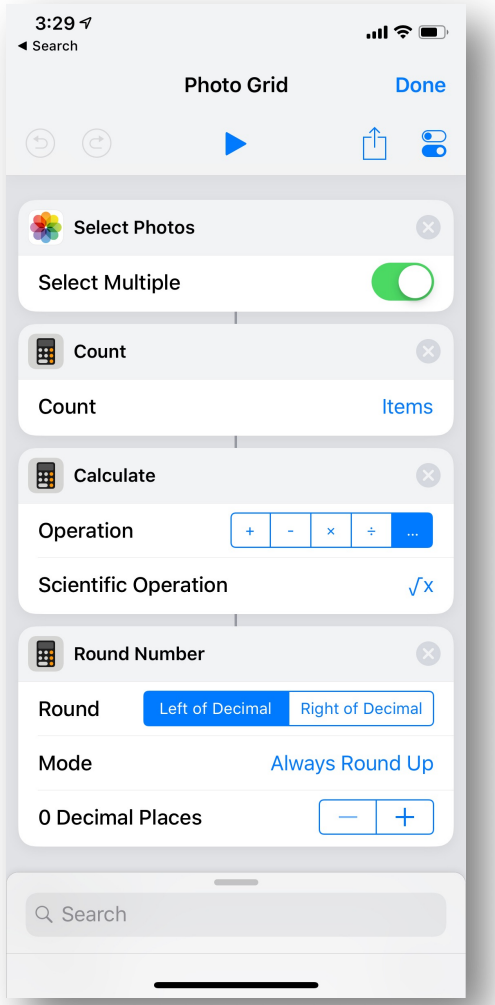
- Proof of concept “Xamarin.Android.Lite”

IOS 12
XCODE 10



Shortcuts – Intents, Siri Integration

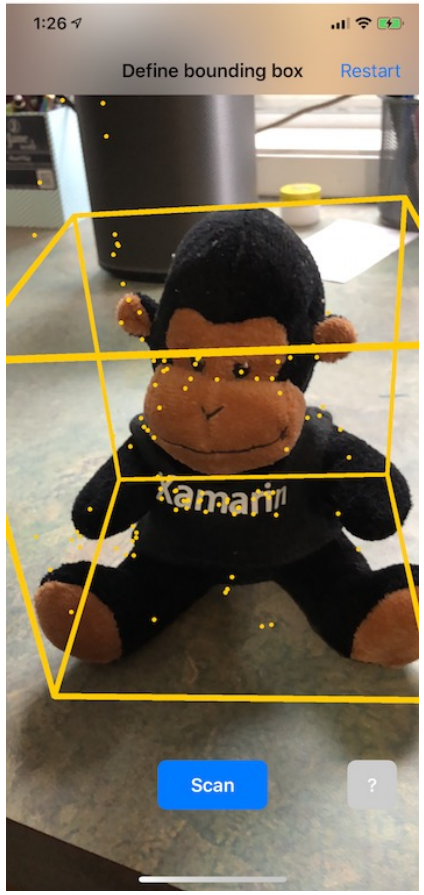


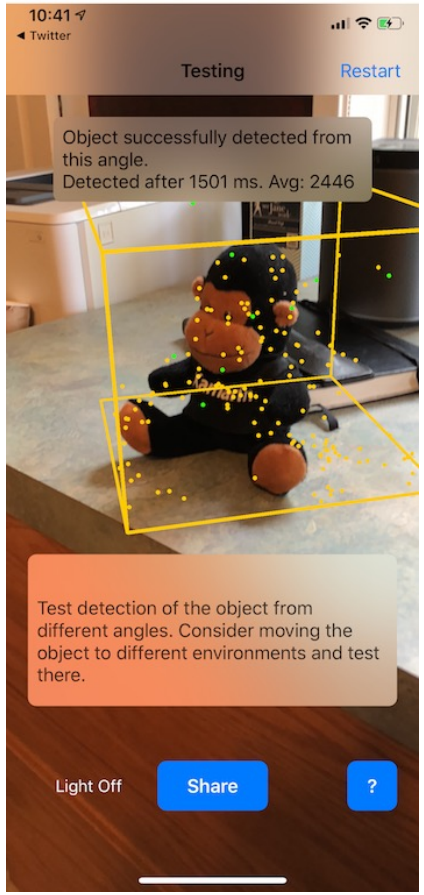


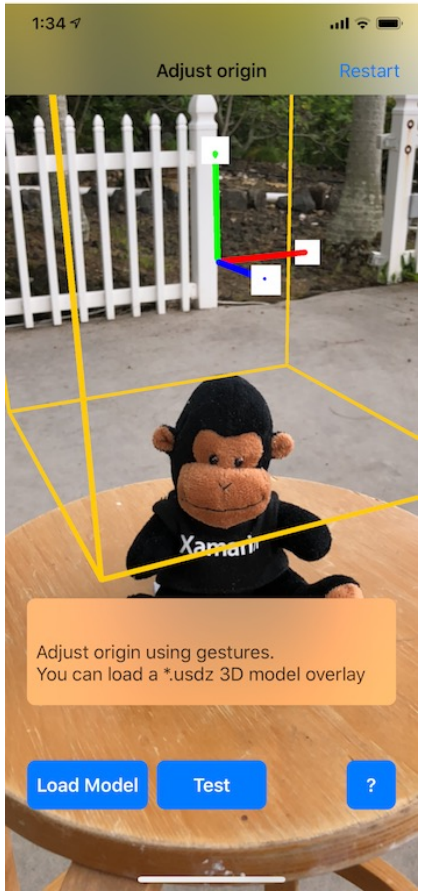
- Surface your app to Siri
- Your app can be part of a script
- App “Donates” actions to the OS, that can be searched, replayed, and can have parameters.

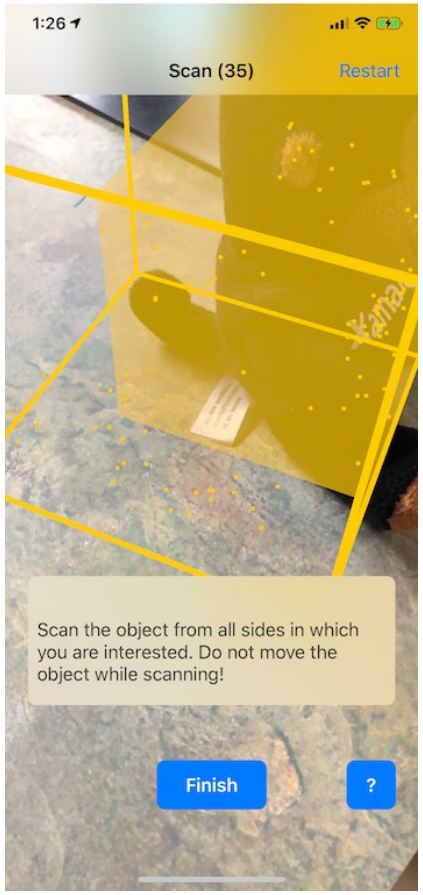
ARKit 2.0

3D object detection, world sharing

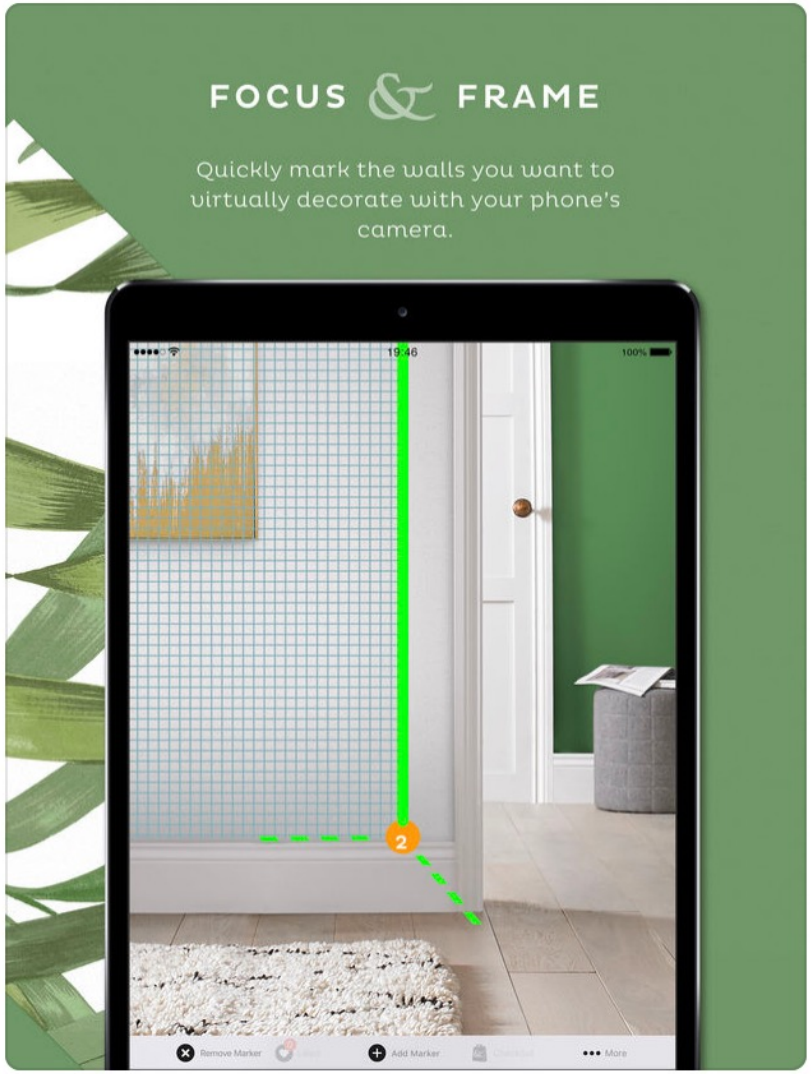


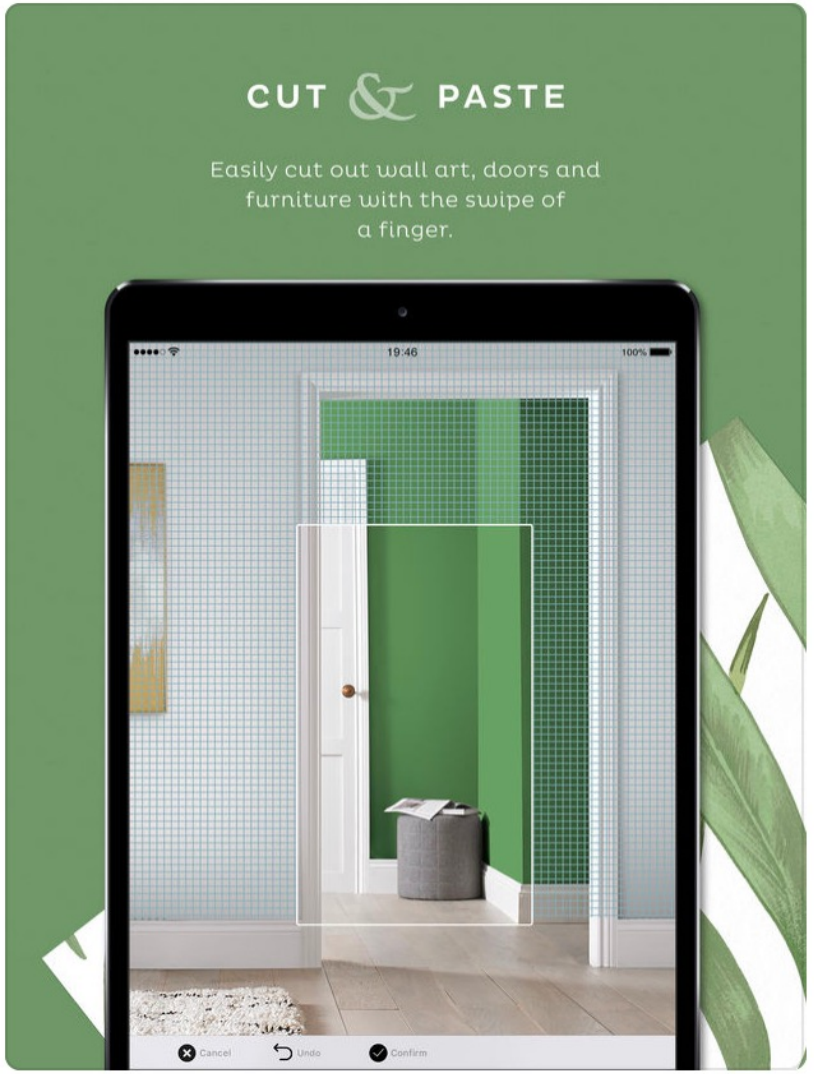


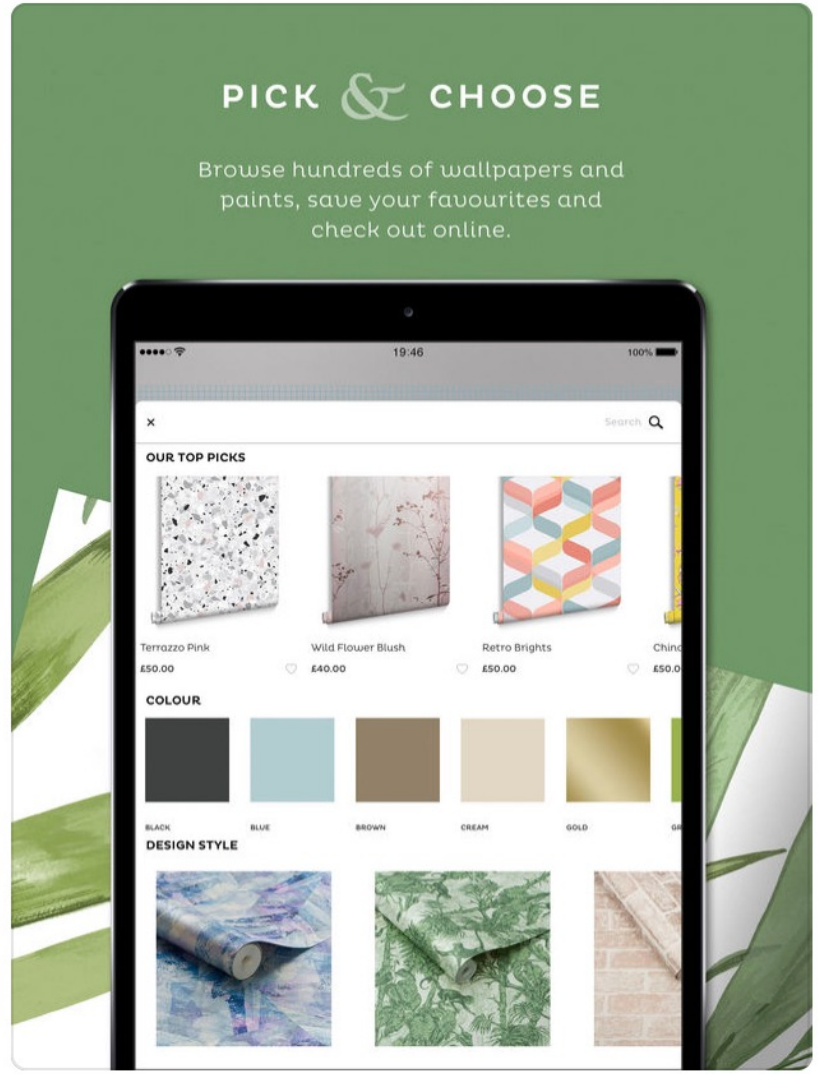




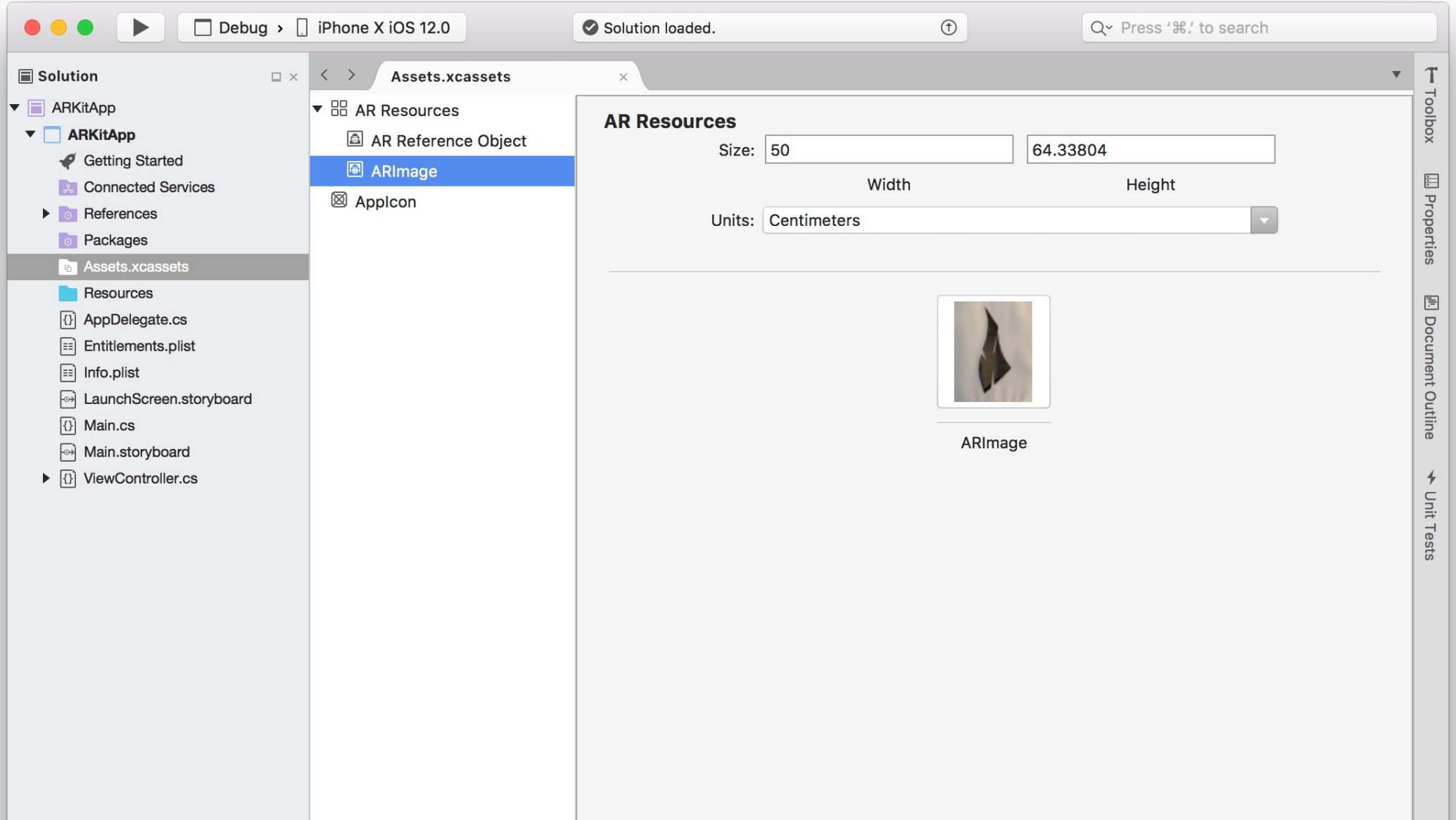
Graham & Brown







ARKit – Resources





XAMARIN.FORMS

LITTLE THINGS

TECHORAMA

DEEP KNOWLEDGE IT CONFERENCE

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Xamarin.Forms Ecosystem















The best UI Toolkit for Xamarin.Forms
Completely redesigned



Designed to make developers' lives easier…

100 +

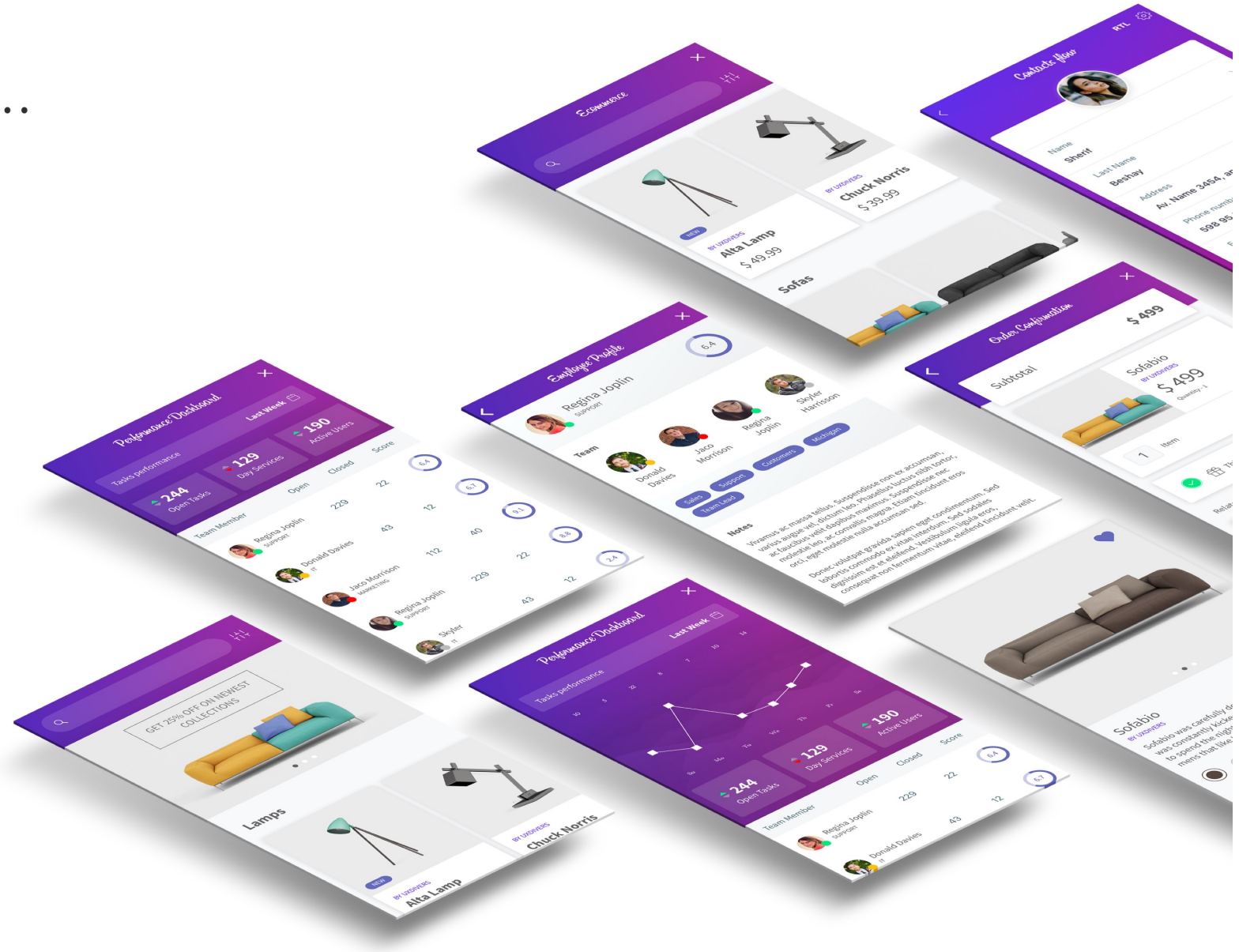
Ready-made XAML pages and templates

15 +

Fully customizable themes

10 +

Custom Controls (DataGrid, VideoPlayer, TabControl, Radios, Checkbox, GrialNavbar and more)





FASTER KICKSTART

Theme Designer
Easy VS Solution Setup



RICHER CONTENT

Fluid Animations, gradients,
custom dialogs & notifications,
charts, and thousands of icons



INSTANT XAML PREVIEW

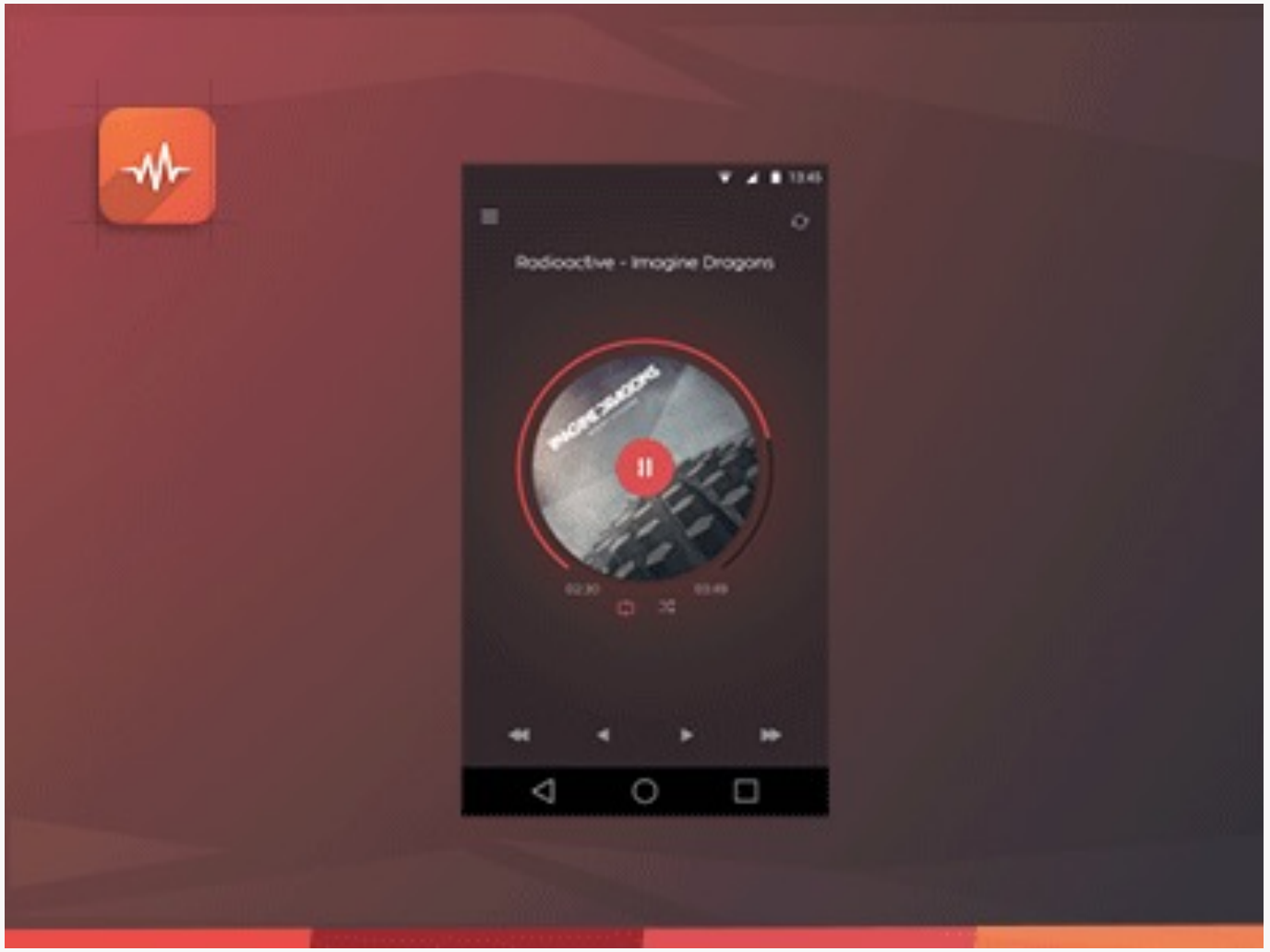
Powered by Gorilla Player

Subscribe and get your discount code at:

grialkit.com/v3



Awesome Xamarin



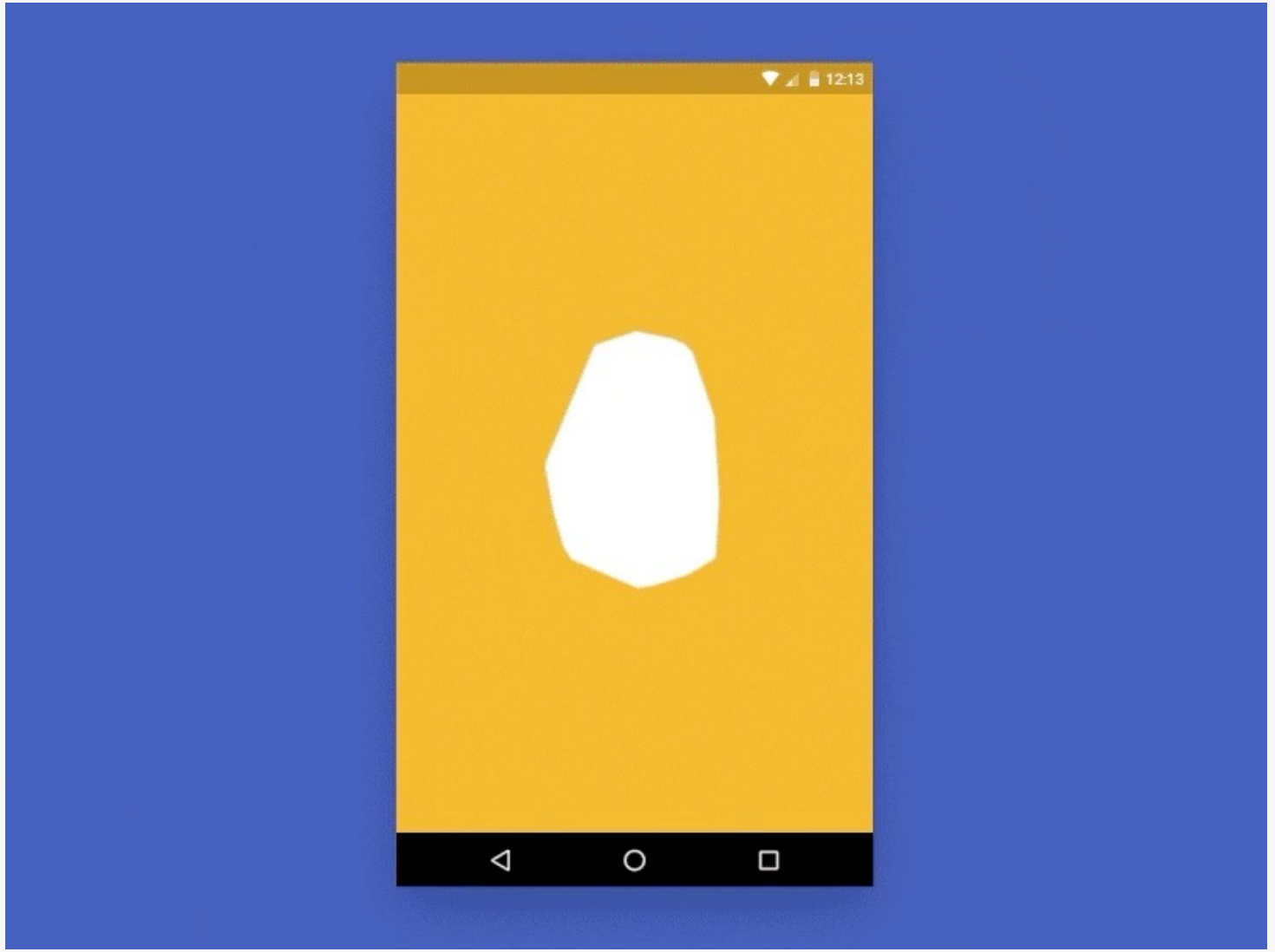
https://javiersuarezruiz.wordpress.com/2018/08/20/xamarin-forms-challenge-pulsemusic/

Awesome Xamarin



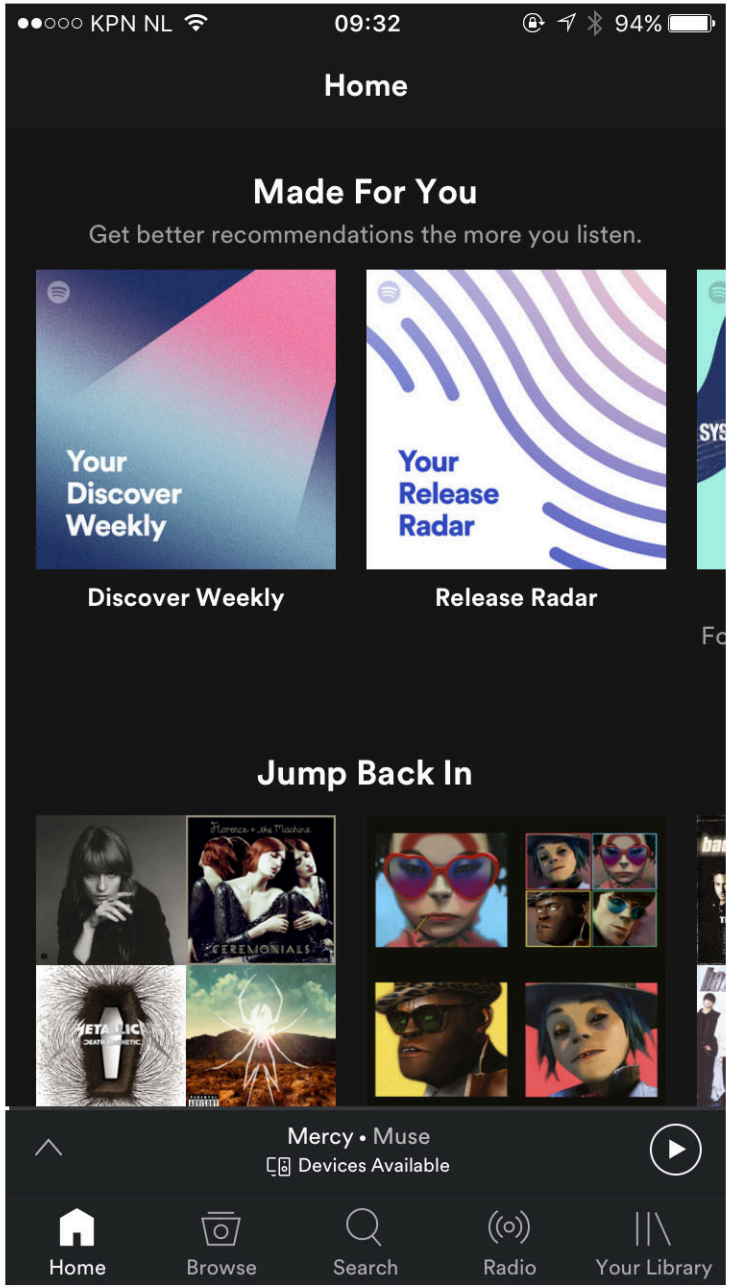
https://github.com/jsuarezruiz/awesome-xamarin-forms

Lottie

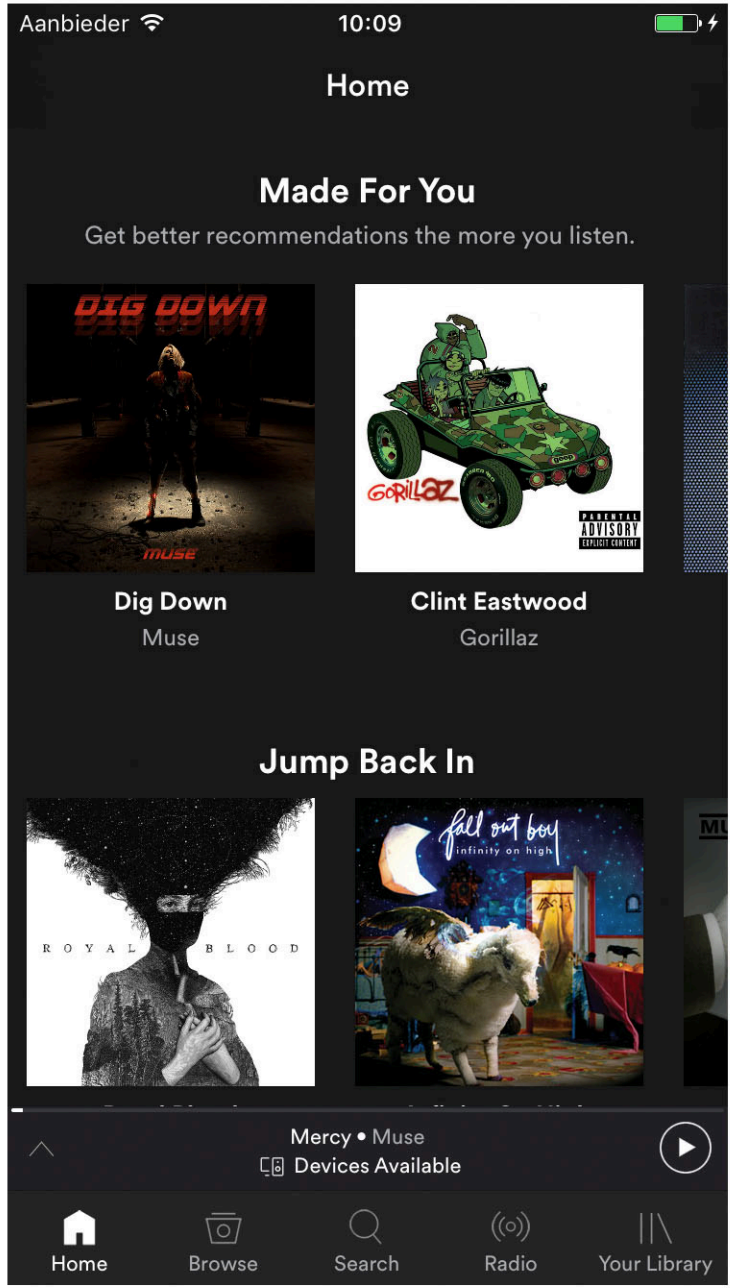


https://t.co/t1DTDGUsv0

Steven Thewissen



The official Spotify app for iOS



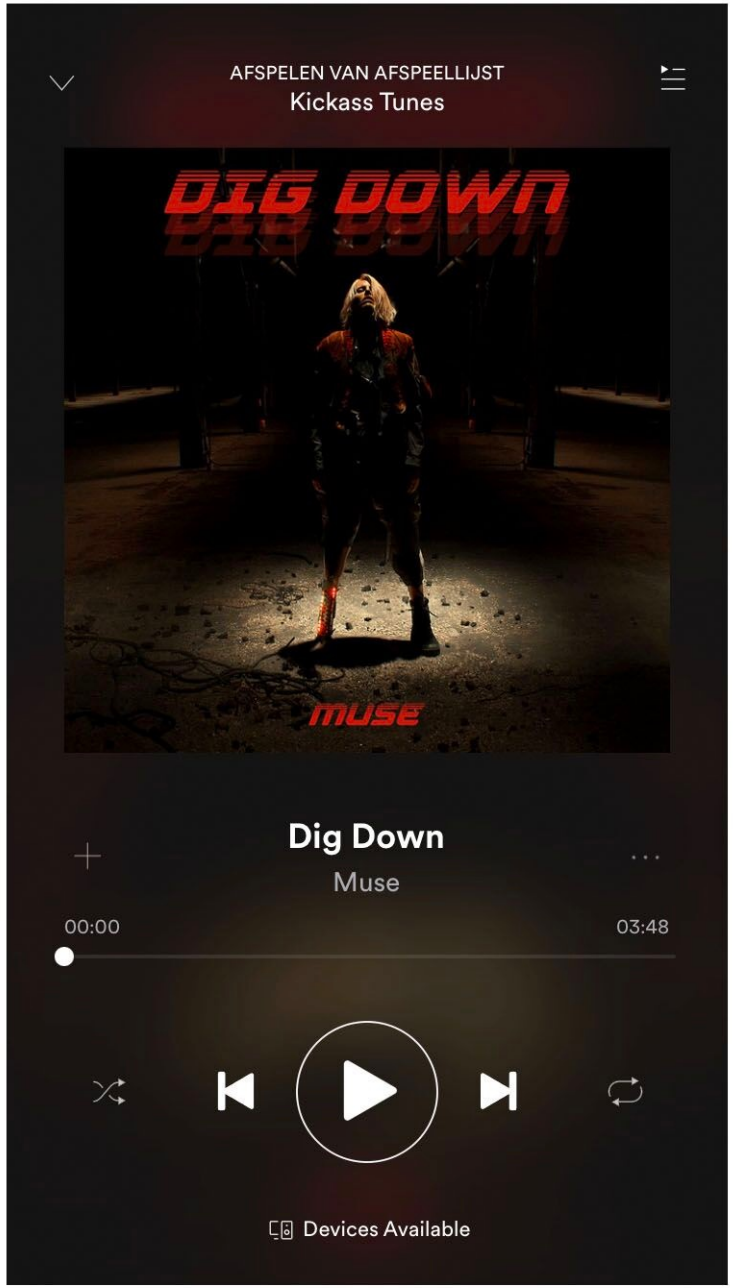
The Xamarin Forms Spotify app for iOS

https://www.thewissen.io/xamarin-forms-ui-spotify/

Steven Thewissen



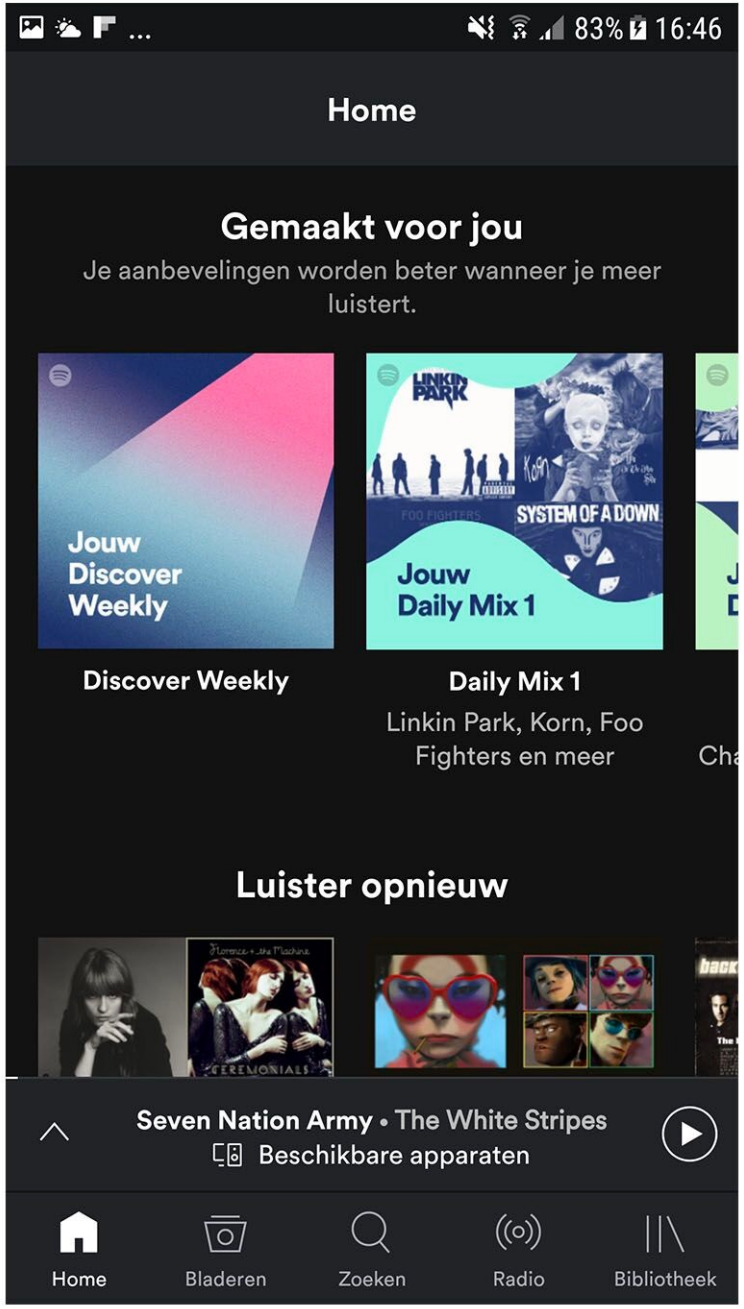
The official Spotify app for iOS



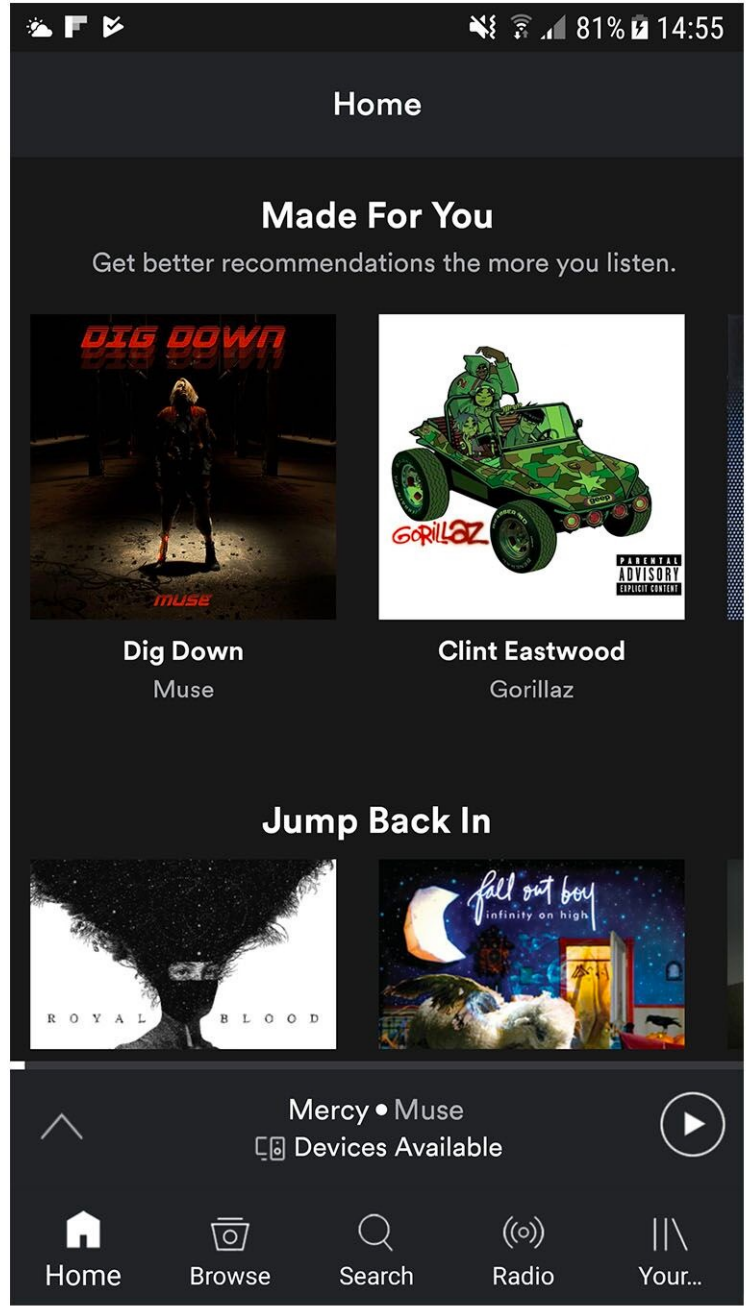
The Xamarin Forms Spotify app for iOS

https://www.thewissen.io/xamarin-forms-ui-spotify/

Steven Thewissen



The official Spotify app for Android



The Xamarin Forms Spotify app for Android

https://www.thewissen.io/xamarin-forms-ui-spotify/

F100: Addressing the Little Things

Project Board

- Entry/Editor
  - IsSpellCheckEnabled
  - MaxLength
  - DetectReadingOrderFromContent (UWP)
  - Text prediction options
  - Autocapitalization
  - AutoResizable Editor
  - ButtonType for return button
  - Position and Color of caret
  - Placeholder color
- Label
  - Bindable Spans
  - Commandable Spans, Span Gestures
  - LineHeight
- ListView
  - Full width separators
  - SelectionMode
- SearchBar IsSpellCheckEnabled
- ImeOptions
- DesignModeEnabled
- Android Bottom Tabs
- WebView mixed content (Android)
- WebView evaluate javascript
- Binding Mode OneTime
- ProgressBar colors
- Switch colors
- Slider colors
- Hide scroll bars
- Button AccessKey
- TabbedPage Icons (UWP)
- Drop Shadows (iOS/UWP)
- App DI Container Hooks
- Button Padding
- Page TitleView
- BoxView rounded corners
- WKWebView for iOS

Coming Soon

Label MaxLines

Entry Read-Only

ImageButton

GIF Support

Button and Label Parity

PinsSource on Forms.Map

WebView Zoom Controls (Android)

Help Wanted

Rounded Corners

Rounded Image

ProgressBar.Height

InkWell Ripple Effect

Size WebView to Content

Floating Action Button

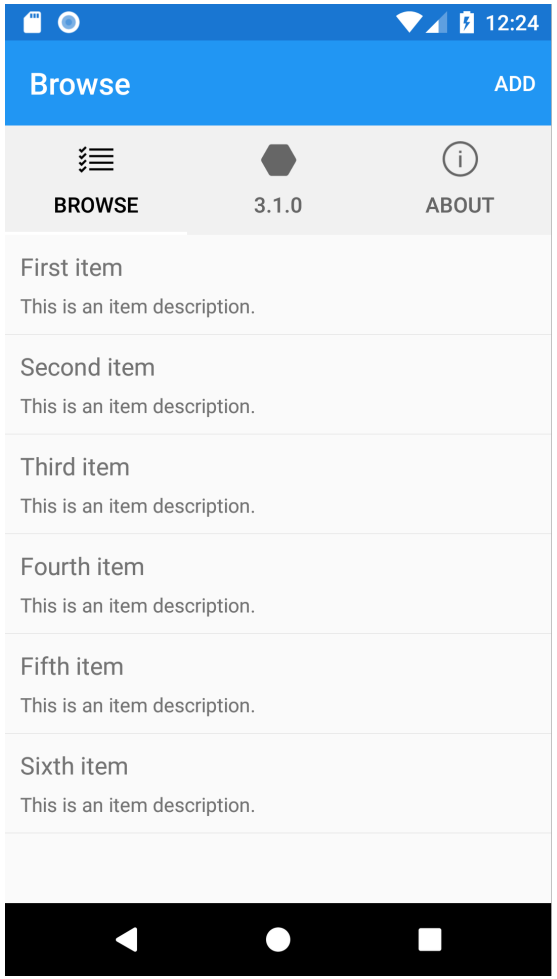
Android Theme Capitalization override

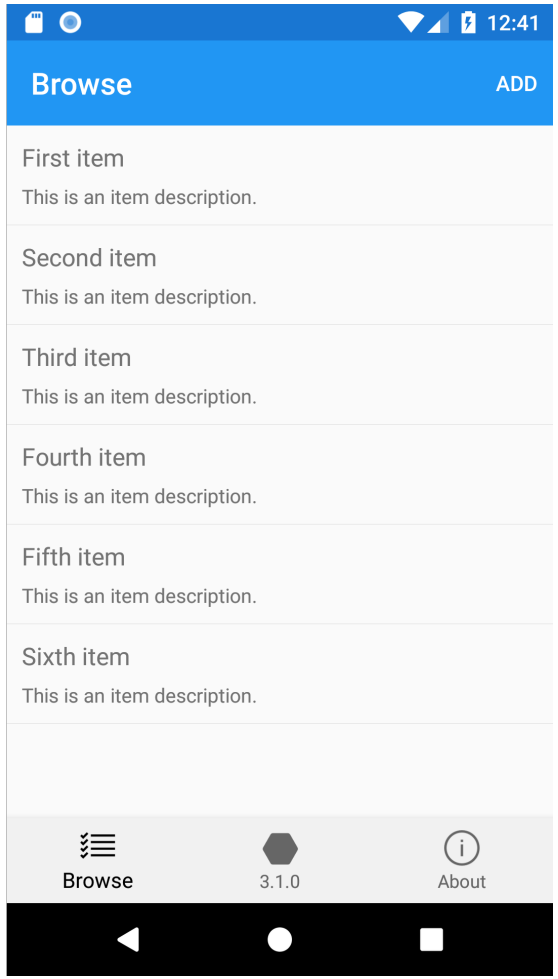
Demo

https://github.com/davidortinau/TheLittleThingsPlayground


LiveXAML
https://www.livexaml.com/
TECHORAMA

Android Bottom Tabs

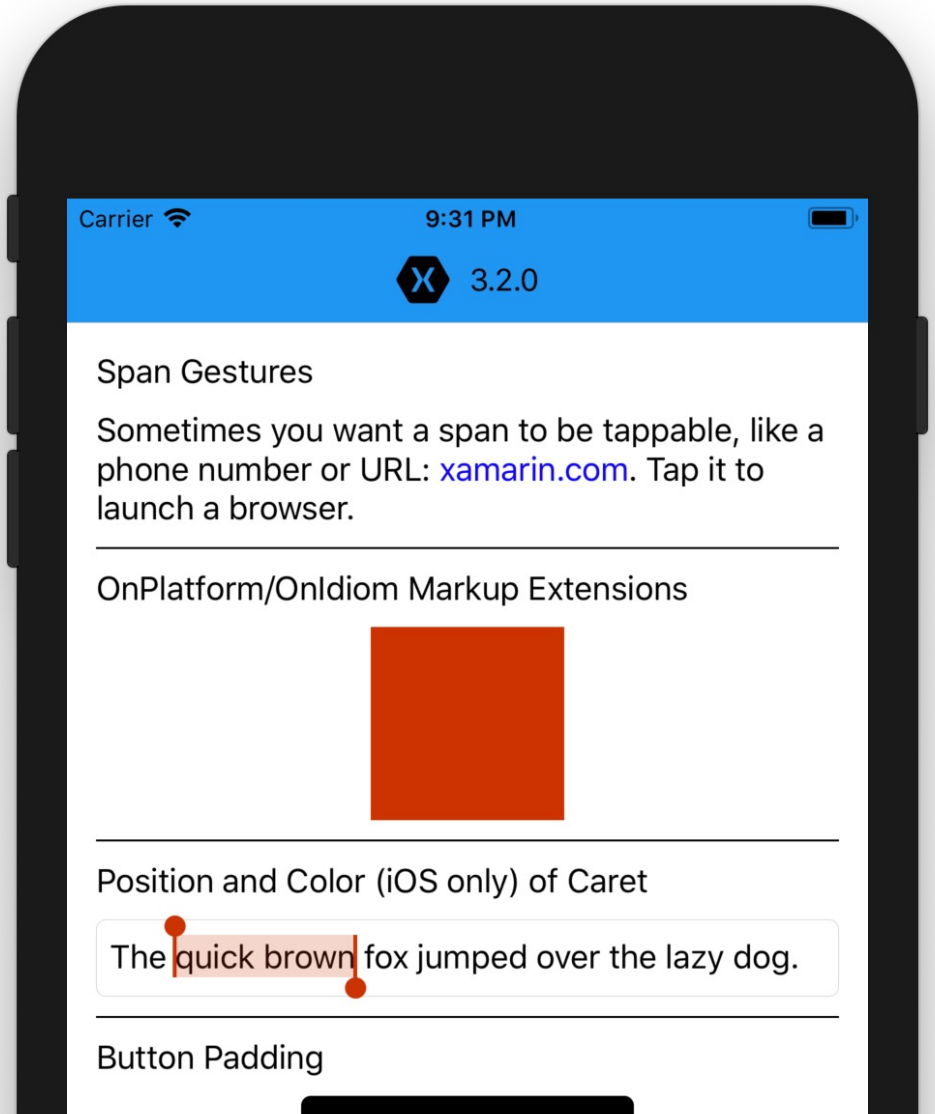





```
<TabbedPage …
            BarBackgroundColor="#F1F1F1"
            xmlns:android="clr-
namespace:Xamarin.Forms.PlatformConfiguration.AndroidSpecific;assembly
=Xamarin.Forms.Core"
            android:TabbedPage.ToolbarPlacement="Bottom"
            android:TabbedPage.BarItemColor="#666666"
            android:TabbedPage.BarSelectedItemColor="Black">

```

TitleView



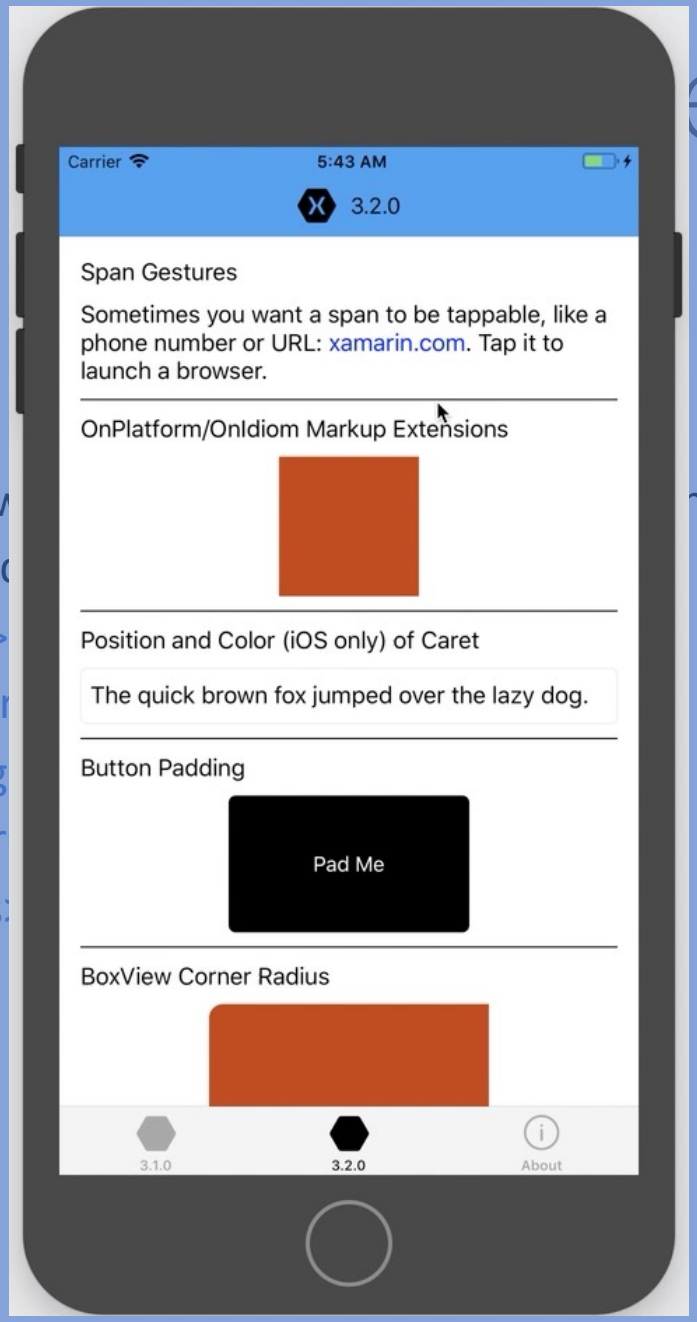

```
<ContentPage>
    <NavigationPage.TitleView>
        <StackLayout
            Orientation="Horizontal"
            VerticalOptions="Center" Spacing="10">
            <Image Source="iconXamagon.png">
                <Image.GestureRecognizers>
                    <TapGestureRecognizer
                        Tapped="Handle_Tapped"/>
                </Image.GestureRecognizers>
            </Image>
            <Label Text="3.2.0"
                FontSize="16"
                TextColor="Black"
                VerticalTextAlignment="Center" />
        </StackLayout>
    </NavigationPage.TitleView>
    ...
</ContentPage>
```

Bindable Span

```
<Label>
   <Label.FormattedText>
      <FormattedString>
         <Span Text="Welcome " />
         <Span Text="{Binding YourName}" FontAttributes="Bold"/>
         <Span Text=" to the Xamarin.Forms 3.1.0 playground!" />
      </FormattedString>
   </Label.FormattedText>
</Label>
```

Commandable es

```
<Label>
    <Label.FormattedText>
      <FormattedString>
        <Span Text="Sometimes you w                                    hone number or URL: " />
        <Span Text="{Binding Url, Mod
          <Span.GestureRecognizers>
            <TapGestureRecognizer
              Command="{Binding
              CommandParameter
          </Span.GestureRecognizers>
        </Span>
        <Span Text=". Tap it to launch
      </FormattedString>
    </Label.FormattedText>
</Label>
```



Button Padding

```
<Button Text="Pad Me"
        Padding="60,40"
        BackgroundColor="Black"
        TextColor="White"/>
```



OnPlatform & OnIdiom XAML Extensions

- Concise syntax
- Supports Default value
- Support Converter and ConverterParamter

```
<BoxView
    FlexLayout.AlignSelf="Center"
    WidthRequest="100"
    HeightRequest="100"
    BackgroundColor="{OnPlatform Android=#333333, iOS=#CC3300, Default=Green}" />
```

Next Things Demos

- CollectionView
- CarouselView
- Shell

Key Features of Shell

- Performant by default
- Same design on iOS and Android via Material Shell or Cupertino Shell
- Platform specific design via Shell
- Single file application structure
- Flyout and Tabbed navigation
- URL routed navigation, deep linking
- Back navigation handling
- Integrated Search handler
- Snackbar
- Bottom Sheet
- Floating Action Button
- Left Bar Button
- Screen segues
- Screen transitions
- Plus everything you can already do today in Xamarin.Forms

THANKS!

david.ortinau@microsoft.com

@davidortinau



Xamarin Essentials

A kit of essential cross platform APIs for your mobile apps – Native Android, Native iOS or Forms



- Accelerometer
- App Information
- Battery State
- Clipboard
- Compass
- Connectivity
- Data Transfer
- Device Information
- File System Helpers
- Email
- Geocoding
- Geolocation
- Gyroscope
- Magnetometer
- Open Browser
- Phone Dialer
- Preferences
- Screen lock
- Secure Storage
- SMS
- Text to Speech
- Version tracking
- Vibrate

More CSS Properties

- ActivityIndicator, BoxView, ProgressBar, Switch
  - color
- Grid
  - row-gap
  - column-gap
- VisualElement
  - transform
  - transform-origin
- Label
  - vertical-align

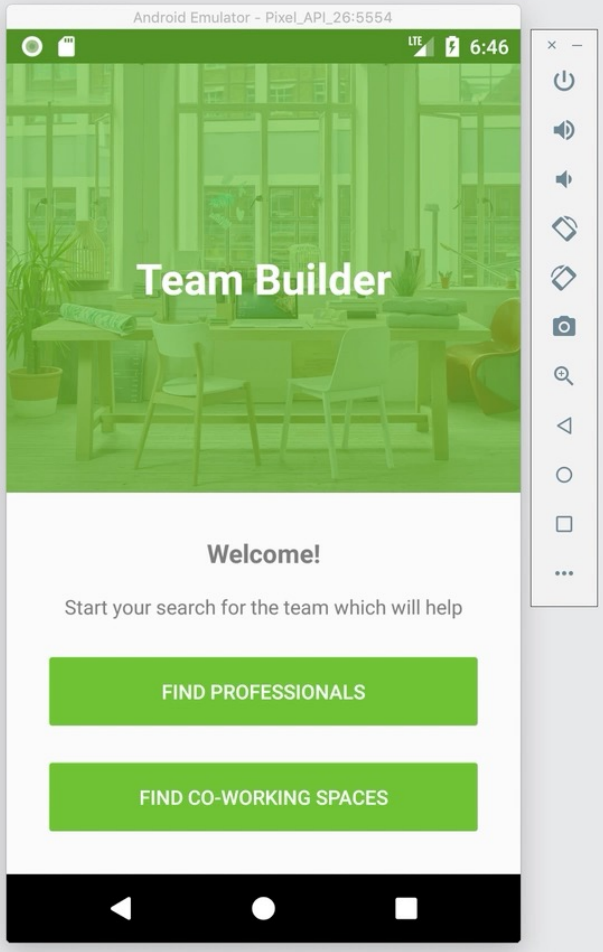
New Xamarin.Forms Specific CSS Properties

- Editor, Entry, SearchBar
  - -xf-placeholder
  - -xf-placeholder-color
- Editor and Entry
  - -xf-max-length
- NavigationPage, TabbedPage
  - -xf-bar-background-color
  - -xf-bar-text-color
- ScrollView and StackLayout
  - -xf-orientation
- ScrollView
  - -xf-horizontal-scroll-bar-visibility
  - -xf-vertical-scroll-bar-visibility
- Slider
  - -xf-min-track-color
  - -xf-max-track-color
  - -xf-thumb-color
- StackLayout
  - -xf-spacing

CollectionView

- ListView but better
- RecyclerView on Android
- You provide the layout
  - Grid
  - Horizontal
  - Vertical
  - FlexLayout (?)
- Context Menus
- Gestures

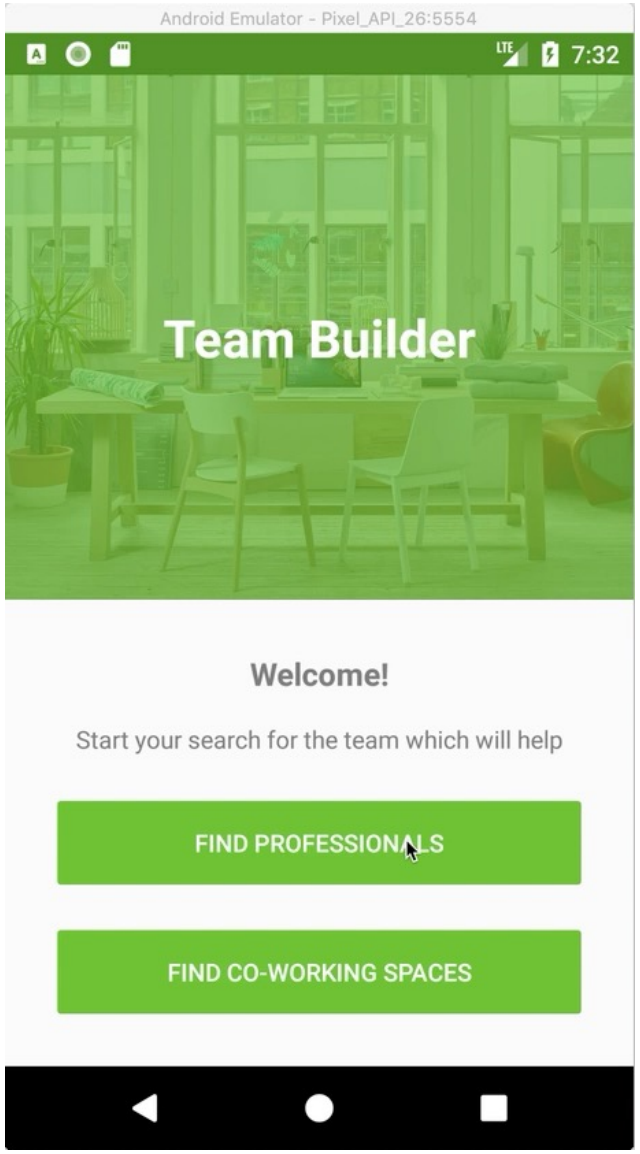
Demo – CollectionView



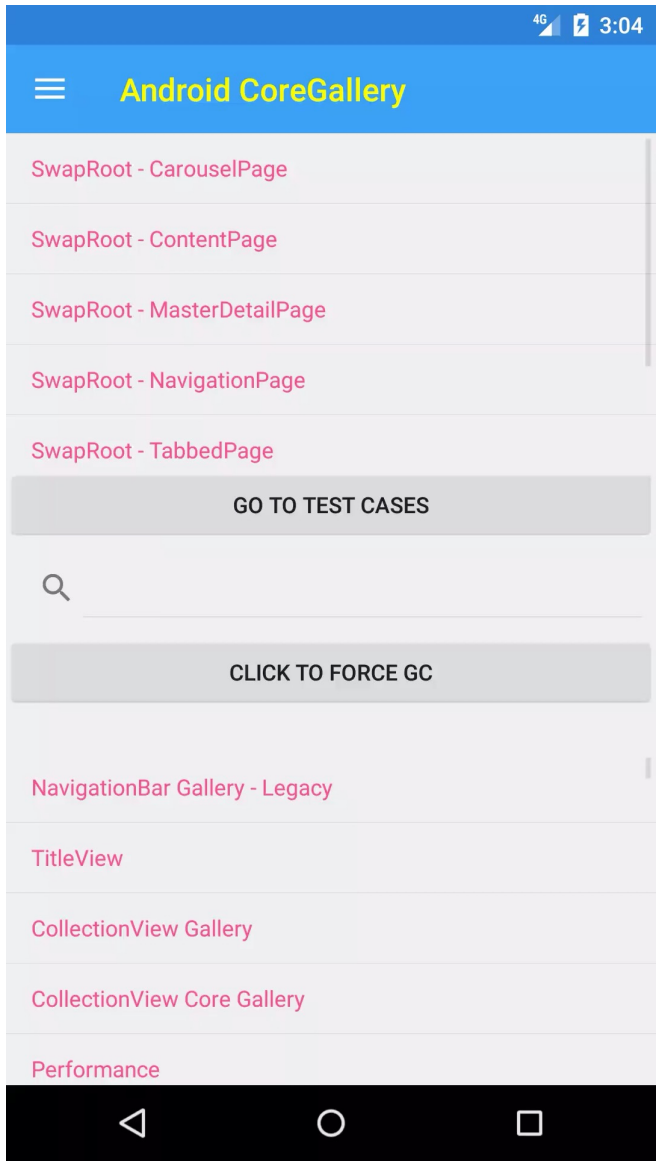
```
<CollectionView x:Name="Spaces" Margin="0,5,0,0”/>

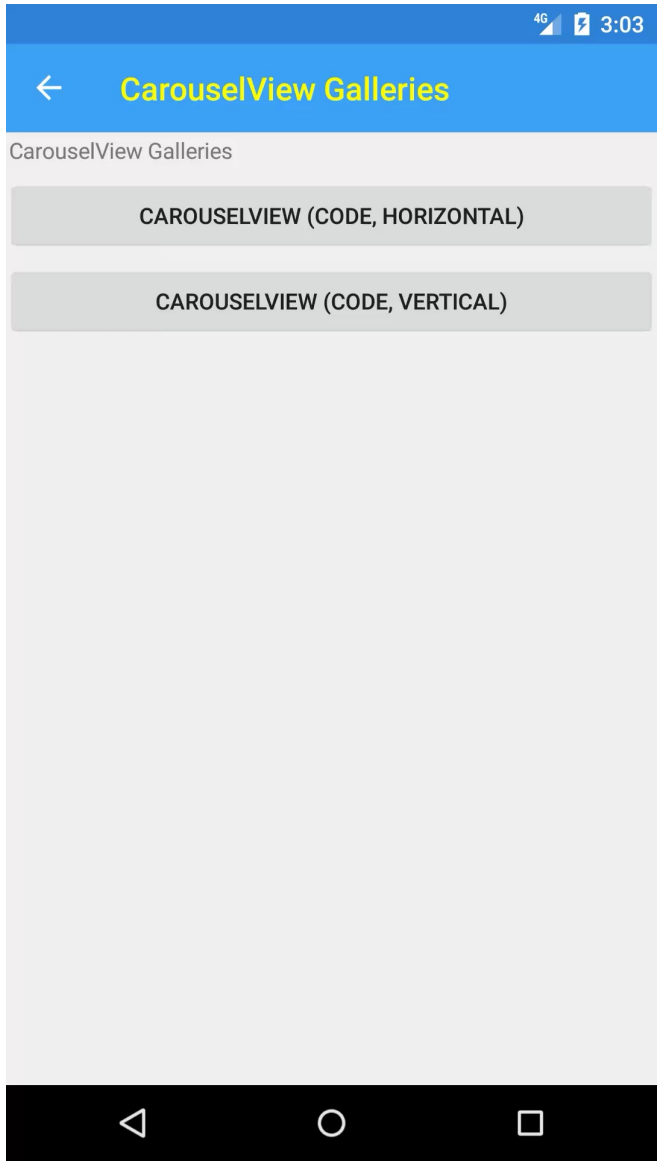
Spaces.ItemsLayout = new GridItemsLayout(2, ItemsLayoutOrientation.Vertical);
Spaces.ItemTemplate = new DataTemplate(() => new SpaceCard());
Spaces.ItemsSource = items;
```

Demo – CarouselView

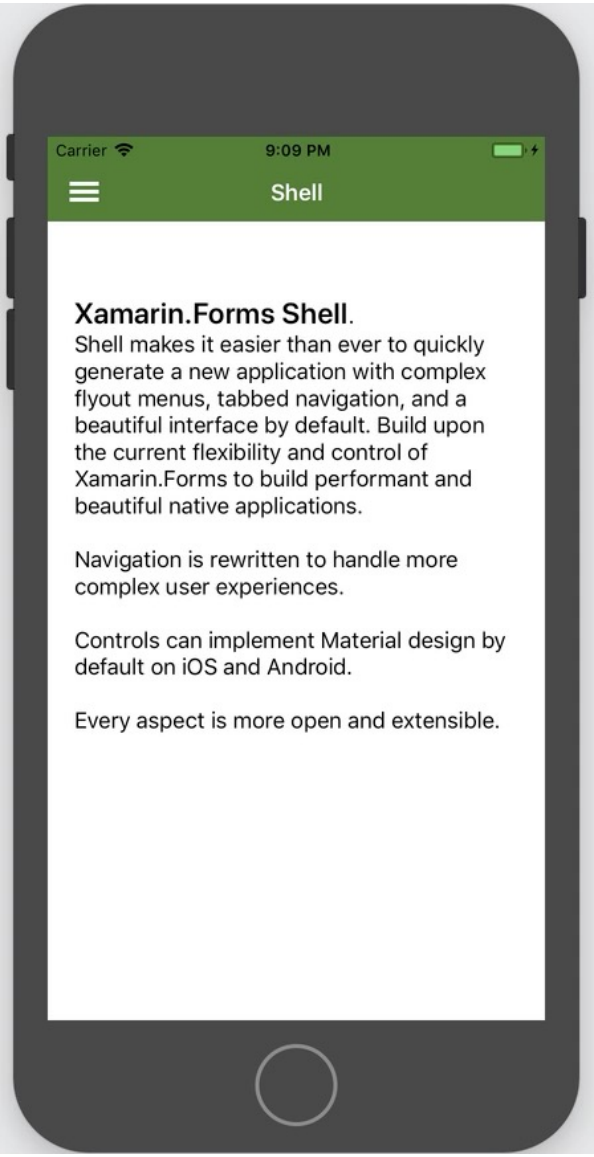


Demo – CarouselView





Xamarin.Forms Shell



- Performant by default
- Same design on iOS and Android via MaterialShell
- Platform design via Shell
- Easily describe your entire app in one file
- Flyout and Tabbed navigation
- URL routed navigation, deep linking
- Back navigation handling
- Integrated Search handler
- Snackbar
- Bottom Sheet
- Floating Action Button
- Left Bar Button
- Screen segues
- Screen transitions
- Plus everything you can already do today in Xamarin.Forms

```
<?xml version="1.0" encoding="UTF-8"?>
<Shell xmlns=…>
    <Shell.Resources>
        <ResourceDictionary Source="Styles/Global.xaml"/>
    </Shell.Resources>

    <ShellItem Route="main" Title="Home">
         <ShellSection Route="home" Title=“Home”>
            <ShellContent ContentTemplate="{DataTemplate local:MainPage}" />
        </ShellSection>
        <ShellSection Route="activity" Title="Activity">
            <ShellContent Route="shared" Title="Shared" ContentTemplate="{DataTemplate local:ActivityPage}" />
            <ShellContent Route="notifications" Title="Notifications" ContentTemplate="{DataTemplate local:NotificationsPa
        </ShellSection>
        <ShellSection Route="updates" Title="Updates">
            <ShellContent ContentTemplate="{DataTemplate local:UpdatesPage}" />
        </ShellSection>
    </ShellItem>
</Shell>
```

```
<Shell.FlyoutHeader>
    <ContentView HeightRequest=“300”>
        <Label Text=“Hello XamStore” …/>
    </ContentView>
</Shell.FlyoutHeader>

<ShellItem Route="main" Title="Home">
    <ShellContent ContentTemplate=“{DataTemplate local:MainPage}” />
</ShellItem>

<ShellItem Route=“notifications” Title=“Notifications”>
    <ShellContent ContentTemplate=“{DataTemplate local:NotificationsPage}” />
</ShellItem>
```

Gestures

- Generic Touch
- Long Press
- Pan
  - IsPanning
- Pinch
- Rotate
- Tap
  - IsTapping
  - NumberOfTouches
  - UpdateEventArgs

Needs Your Input and Vote

https://github.com/xamarin/Xamarin.Forms/labels/a%2Fgestures

THANKS!

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